

An Interpretable Hybrid Machine Learning Framework for Climate Air Quality Dynamics

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ABSTRACT

Predicting rainfall and temperature based on air quality indicators is considered one of the more difficult tasks. Complex interactions between air masses, nonlinear and interdependent connections, and overfitting problems are related to this subject. In response to these challenges, this study introduces the Air Quality-based Regularized Optimized Prediction Model (AQ-ROMP), an interpretable hybrid machine learning model that combines exploratory statistical analysis, Lasso-based feature selection, Extreme Gradient Boosting (XGBoost) regression, and Bayesian hyperparameter optimization into a single prediction pipeline. To test the framework, the Indian Climate Dataset (2024–2025) was used, which includes N environmental observations from M monitoring locations, including air-quality indicators (PM_{2.5}, PM₁₀, CO, NO₂, SO₂, O₃, and AQI) and meteorological variables. Before training the models, the dataset was preprocessed by removing outliers, imputing missing values, normalizing the features, and performing Lasso feature selection. The performance was evaluated by 80:20 train–test split with 5-fold cross validation on the training set and evaluated by MAE, RMSE, R², MAPE and EVS. In the experiments, AQ-ROMP demonstrated excellent predictive performance, with R² values of about 0.94 for temperature and 0.91 for rainfall, thereby surpassing traditional methods by 8-15%. The model developed, however, is prone to seasonality and data interdependence. In conclusion, the proposed architecture is a robust, scalable model for weather prediction and ecologically friendly decision-making.

Keywords: *Air Quality Parameters, Temperature and Rainfall Prediction, Model Regularization, XGBoost, Bayesian Optimization, Climate Forecasting.*

1. INTRODUCTION

The interaction between air pollution and climate change has negative impacts on human health, agricultural productivity, and even ecological stability [1], and is increasingly complex due to numerous interrelated factors. Levels of particulate matter (PM_{2.5}, PM₁₀), carbon monoxide (CO), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), and ozone (O₃) have increased due to rising vehicle emissions, industrial growth, and urbanization [2]. Two of the numerous negative effects of air pollution are deterioration of air quality and alteration of weather patterns due to changes in atmospheric processes [3]. Only with this knowledge can we design accurate climate-monitoring systems, disaster-preparedness plans, and environmentally friendly policies [4]. As the quantity of large-scale environmental datasets has increased [5], it has become increasingly accessible to construct sophisticated predictive models that can represent these complex relationships [6].

Traditionally, statistical techniques, including Multiple Linear Regression (MLR) and other empirical models, have become a common tool for predicting environmental factors [7]. These techniques have significant disadvantages, although they are easy to use and interpret. Atmospherics are inherently complex [8], with nonlinear relationships and interactions among many variables that are hard to represent well with traditional linear models [9]. The traditional techniques are also characterized by severe multicollinearity and an inability to handle high-dimensional data [10]. The methods are already having difficulty generalizing to new data and are much less reliable due to overfitting and underfitting [11]. Modern methods also do not adapt well to changing seasons or other environmental conditions, and they do not predict future climates with great accuracy [12]. These issues mean that the air quality variables cannot be included in a weather forecast system without a more comprehensive model [13]. The presence of many environmental factors, nonlinear relationships, and high-dimensional input data are reasons why problems should be solved [14]. To make the prediction algorithm more complex and avoid overfitting, regularization methods are required. Optimization comes into its own in finding the best values, especially those of the model's parameters, making it a necessity for achieving optimal performance [15].

Traditional approaches are used to make models more accurate and reliable. ML is an experience-based algorithm for model creation that uses historical data to train the model [16]. It is possible to create highly customized models using some of the methods. By adding a decision tree at each iteration, Extreme Gradient Boosting (XGBoost) enhances performance and is an example of ensemble learning [17]. To further aid in complexity management and enable the

selection of the most relevant predictors, regularization penalizes specific model parameters [18]. Optimal performance can also be achieved through hyperparameter optimization, which systematically seeks the best values for hyperparameters [19]. It is possible to measure the efficacy of ML models using various metrics. Two ways to quantify the magnitude of the prediction error are root mean square error (RMSE) and mean absolute error (MAE).

In contrast, the R^2 stands for the capacity to account for data variability. After everything is said and done, these approaches and standards can be used to create scalable models that accurately predict weather conditions, such as temperature and rainfall, using air quality indicators [20, 21]. Decisions about environmental management can be better informed with this information. The paper has made key contributions, which include,

- We suggest an end-to-end interpretable framework AQ-ROPM which is systematically developed to combine exploratory statistical analysis, Lasso based regularized feature selection, nonlinear regression based on XGBoost and Bayesian hyperparameter optimization into a single prediction pipeline.
- Compared to the traditional studies that apply these techniques separately, AQ-ROPM provides a progressive learning process: statistical dependency analysis is employed to regularize the features in the feature space, the nonlinear model construction is designed to improve the interpretability and complexity of the model.
- The proposed framework makes the prediction of temperature and rainfall based on air quality parameters simultaneously, while most of the earlier studies dealt with the prediction of only one of the variables like air quality index or the prediction of PM2.5.
- AQ-ROPM presents an interpretable feature refinement strategy to remove redundant environmental variables from the input prior to nonlinear learning, hence reducing overfitting without compromising the predictive ability.
- An extensive ablation study quantitatively proves the individual contribution of feature regularization, optimization and air-quality information and validates the effectiveness of the whole process.

Problem Statement

The prediction of temperature and rainfall based on air quality measurements can be modeled as a supervised learning problem with multiple inputs and outputs. Within the input feature set $X = \{x_1, x_2, \dots, x_n\}$, each x_i is related to air quality and other atmosphere-related factors, including particulate matter (PM2.5, PM10), carbon monoxide (CO), nitrogen dioxide (NO2), sulfur dioxide (SO2), ozone (O3), and other additional environmental parameters. The variables that are to be

measured are $Y = \{T, R\}$, that is, temperature (T) and rainfall (R). Researching a mapping function $f: X \rightarrow Y$ that can forecast T and R given inputs is the objective of the task. Since the input features are too many and may overlap, regularization is necessary to minimize overfitting and isolate the most important variables. By including a penalty term in the loss function, we can do this and guarantee that the model is both simple and predictive. Adjusted to perform better are also the hyperparameters that control the learning process. The overall aim is to enhance accuracy and generalizability: to maximize the coefficient of determination (R^2) and reduce prediction error (quantified by MAE and RMSE).

The proposed AQ-ROM framework is designed to solve a dual-target prediction problem, where the air-quality and meteorological predictor variables are the same. Let X be the input feature matrix, which includes PM2.5, PM10, CO, NO₂, SO₂, O₃, humidity, pressure, wind speed, cloud cover, and other environmental factors. The prediction task is predicting two target variables – temperature (Y_1) and rainfall (Y_2). The two prediction tasks have the same preprocessing pipeline, feature selection procedure and hyperparameter optimization strategy, but two separate XGBoost regression models are trained for temperature prediction and rainfall prediction. This design allows each target variable to find its own nonlinear relationship between the environmental predictors, while keeping the framework of the AQ-ROM simple and interpretable.

2. RELATED WORKS

The Pak et al. [22] air quality prediction model contains the Least Absolute Shrinkage and Selection Operator (LASSO) to enhance feature selection and minimize overfitting. Data from 16 sensors is used to test ML models. The reason is that measures such as R-squared, MAE, MSE, RMSE, and NMSE are used. The results further indicate that particulate matter (PM2.5, PM10) can be better predicted than gases (CO, NO₂, SO₂, O₃). Two of the weaknesses are the variability of gases and the limitations of models.

Suthar et al. [23] have proposed a land surface temperature (LST) forecasting model for Bengaluru based on ML models that combine air pollutants, urbanization variables, and surface indices with meteorological data. Artificial neural networks (ANNs), multiple linear regression (MLR), support vector regression (SVR), and random forests (RFs) are also available. ANN achieved superior performance ($R^2 = 0.92$ in summer, 0.95 in winter; lower RMSE and MSE). Limitations include seasonal variability and complex pollutant interactions, which affect generalization.

Natarajan et al. [24] proposed a methodology for predicting the Air Quality Index (AQI) that combines Grey Wolf Optimization (GWO) with Decision Tree (DT) to improve accuracy. It evaluates k-nearest neighbor (KNN), RF, and SVR using air pollution data from Kaggle. The

results demonstrate superior performance (accuracy up to 97.68%). Relying on a single dataset and potential problems with generalization across different environments are two limitations.

Manna, T., & Anitha, A. [25] proposed REG-CLSTM, a regularized combination of long short-term memory (LSTM) for seasonal air quality prediction. Rough set-wrapper feature selection and hybrid deep learning based on pyramid learning are two techniques that are used. The model identifies seasonal limitations in pollutant levels and improves accuracy, speed, and error rates. Complexity, high computational expense, and reliance on massive real-time datasets limit generalizability and scalability.

Mampitiya et al. [26] proposed a method for predicting PM10 levels in Sri Lanka using ML and DL models. The methods being assessed include XGBoost, CatBoost, Light Gradient Boosting Machine (LightGBM), LSTM, GRU, R³, RMSE, MAE, MSE, MARE, and Nash-Sutcliffe efficiency (NSE). By comparing the two methods, LightGBM proves to be more accurate ($R^2 = 0.99$). Generalization is affected by limitations such as geographical dependence and seasonal variations.

Waseem et al. [27] proposed that Pakistan could benefit from PM2.5 forecasts powered by data-driven ML and deep learning models. Here are some techniques for analyzing weather impacts: FbProphet, LSTM, and LSTM encoder-decoder. According to the results, the LSTM encoder-decoder outperforms the others, achieving an hourly MAPE of 7.4%. The reliance on meteorological factors that affect prediction consistency and the substantial error variability across cities are limitations.

Masood et al. [28] proposed a model for PM2.5 forecasting that combines extreme learning with snake optimization (ELM-SO). Methods include ELM-SO, XGBoost, LSTM, SVR, RF, gradient boosting regressor (GBR), and several others. Superior performance is demonstrated by the results ($R^2 = 0.928$, RMSE = 30.325 $\mu\text{g}/\text{m}^3$). Possible limitations include problems with generalization across different situations and reliance on the quality of input.

Ravindiran et al. [29] suggested a method to forecast the Air Quality Index (AQI) for Chennai utilizing machine learning (ML) models and data from 2017 to 2022. Log transformation is used to normalize the results of XGBoost, Random Forest (RF), BaggingRegressor, and LGBMRegressor. $R^2 = 0.9935$, MAE = 0.02, MSE = 0.001, RMSE = 0.04, the results demonstrate that XGBoost delivers better performance. Some limitations include difficulty adapting to new environmental conditions quickly and excessive reliance on past trends.

In recent years, feature selection methods [31] have been emphasized for enhancing the robustness, interpretability, and generalization of machine learning models. Barbieri et al.

extensively reviewed some of the most popular feature selection methods and showed that the selection of features significantly reduces the dimensionality of the data, removes redundant variables, makes the prediction more stable and makes the model more interpretable in various machine learning applications.

However, some challenges still need to be considered. First of all, there exists the issue of single-target prediction only. Current literature is mostly focused on predicting air quality metrics such as PM_{2.5} and the Air Quality Index (AQI). Multi-target prediction that considers such meteorological factors as rainfall and temperature does not attract much attention. In addition, there is an obvious focus on improving accuracy rather than introducing the regularizations necessary to reduce overfitting in high-dimensional datasets. Air quality parameters and meteorological factors are not necessarily considered together in modeling. Moreover, most models do not pass cross-regional validation, which may reduce their applicability. Finally, there is no attempt to consider both machine learning and deep learning algorithms simultaneously, even though this approach may yield better results. The seasonality effect also remains outside the scope of most climate prediction studies. Improving climate prediction frameworks in terms of reliability, scalability, and generalizability requires filling these gaps.

The novelty of the proposed AQ-ROM framework lies in its integrated methodological design rather than in the development of new standalone machine learning algorithms. Lasso regularization, XGBoost and Bayesian optimization are all well-known methods, but previous studies on environmental prediction problems typically use these methods in isolation or only for a single-target forecasting problem, e.g., predicting the air quality index or the concentration of a pollutant. AQ-ROM, on the other hand, introduces a unified end-to-end solution for simultaneously forecasting temperature and rainfall by combining statistical feature analysis, regularized feature selection, nonlinear ensemble learning, and Bayesian hyperparameter optimization. Moreover, the proposed framework integrates interpretable feature selection, optimized nonlinear prediction and extensive ablation analysis, which offers an effective and transparent solution for multi-target climate prediction.

3. AIR QUALITY-DRIVEN REGULARIZED OPTIMIZED PREDICTION MODEL (AQ-ROM)

The proposed AQ-ROM model begins with aggregating historical environmental data, including air quality metrics such as PM_{2.5}, PM₁₀, CO, NO₂, SO₂, and O₃, along with relevant temperature and rainfall records. The X is the input, and $Y = \{T, R\}$ is a target variable. Based on Figure 1, the proposed model includes preprocessing, exploratory data analysis, feature selection via Lasso regularization, nonlinear modeling with the XGBoost algorithm, and Bayesian optimization to

produce accurate and robust multi-output weather forecasts. Firstly, an exploratory analysis is performed better to understand the distributions of each feature and their correlations. The feature selection is then performed using Lasso regularization to select the most relevant features and ignore redundant ones, thereby minimizing model dimensionality and overfitting. After feature selection, a reduced dataset is developed to train a predictive model based on extreme gradient boosting (XGBoost), which effectively models the interactions and nonlinearity among meteorological variables. A subset of Bayesian optimization is hyperparameter tuning, which aims to optimize a model's performance by selecting the best hyperparameter configuration.

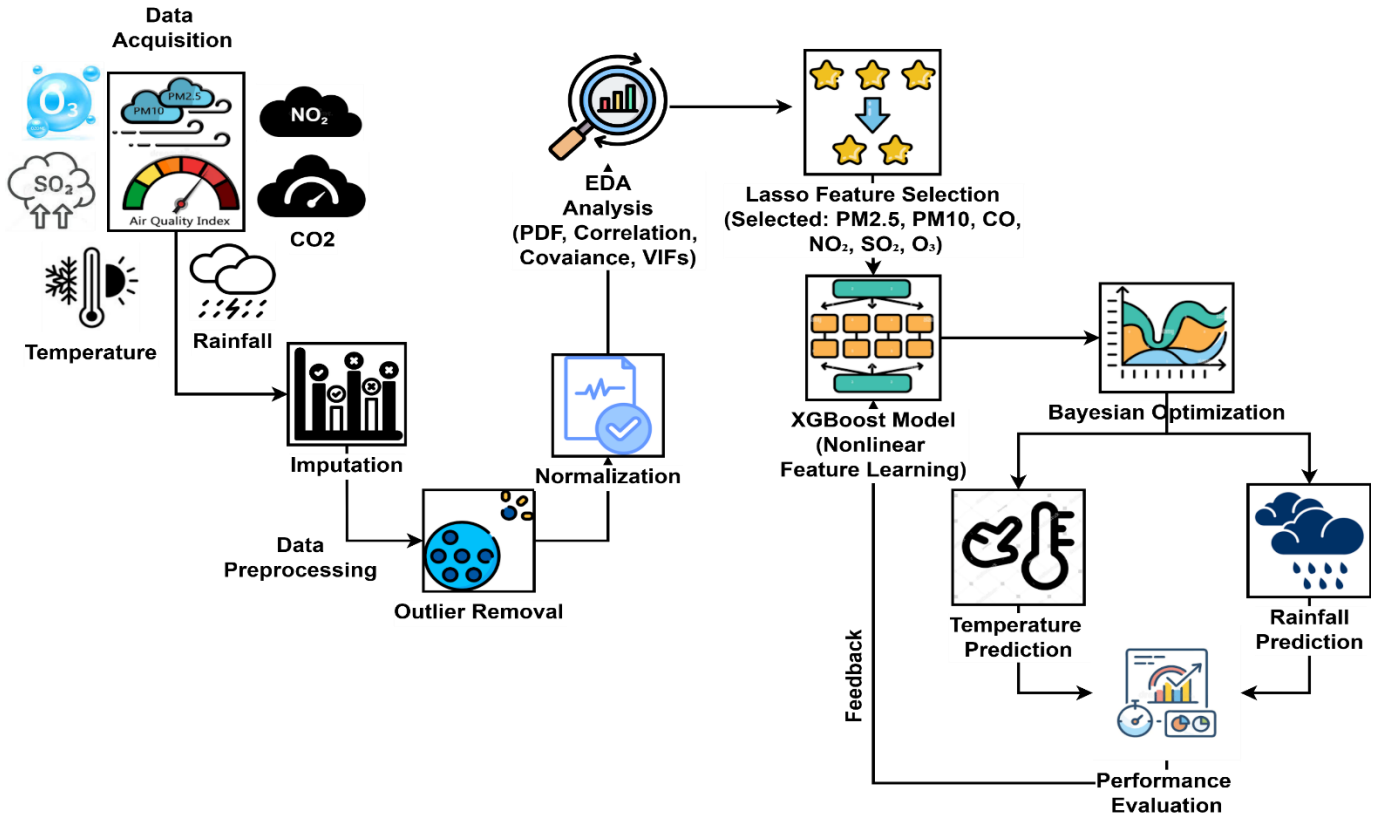


Figure 1: Overall architecture of the proposed AQ-ROPM framework for temperature and rainfall prediction.

Optimizing the model parameters by training on the best parameters, and evaluating the model on new data to measure generalization. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2) are some of these performance measures. To ensure the predictions are uniform and reliable, we verify the forecasted temperature and rainfall. The characteristics of this methodical approach include improved accuracy, reduced overfitting, and robust weather prediction enabled by air quality measures.

3.1 Dataset Description

The Indian Climate Dataset (2024-2025) [30] from Kaggle was used to assess the proposed AQ-ROM framework. The data includes meteorological data as well as air quality measurements at various monitoring stations in India. The observations are each a daily environmental record, yielding synoptic observation of the pollutants and other weather variables for supervised regression analysis. The entire dataset comprises of 7000 environmental observations taken from various locations across India with varying climatic conditions. The data in each record includes PM2.5, PM10, CO, NO₂, SO₂, O₃, Air Quality Index (AQI), temperature, rainfall, humidity, air pressure, wind speed, and cloud cover, as well as temporal data which maintains the temporal continuity. The daily temporal resolution allows the proposed framework to be able to reflect the relationship between air quality variations and short term changes in meteorological conditions. The data set was reviewed for data quality and consistency before developing the model. A small percentage of missing observations (less than 3% of the total observations) was found. In the case of numerical attributes with missing values, the data were imputed using mean imputation for those that were approximately normally distributed, and median imputation for the skewed pollutant measurements. Z-score method ($|Z| > 3$) was applied to detect outliers, and they were deleted prior to the training of the models. Then, all numerical features were scaled to the same range (Min-Max scaling) for model learning. The preprocessed data set was split randomly into 80% training data (5600 samples) and 20% independent test data (1400 samples). Thereafter, five-fold cross validation was carried out in the training subset for model validation and the optimization of the model's hyperparameters by the method of Bayesian. The statistical properties of the environmental variables used in the present study are shown in Table 1.

Descriptive statistical characteristics of all predictor and target variables used in this study are presented in table 1. The reported values such as minimum, maximum, mean and standard deviation were calculated directly from the preprocessed dataset used for the model development and evaluation. These statistics are a summary of the distribution of the environmental and meteorological variables, and will form the foundation for the feature selection and predictive modelling processes that follow.

Table 1: Statistical summary of air quality and meteorological variables used in the study.

Variable Name	Min Value	Max Value	Mean Value	Unit
PM2.5	10	350	95	µg/m ³
PM10	20	450	140	µg/m ³
CO	0.2	3.5	1.2	mg/m ³
NO ₂	5	180	60	µg/m ³

SO₂	2	90	25	μg/m ³
O₃	10	200	70	μg/m ³
AQI	50	400	180	Index
Temperature	18	42	30	°C
Rainfall	0	250	75	mm
Humidity	30	95	65	%
Pressure	980	1035	1010	hPa
Wind Speed	0.5	15	5.5	m/s
Cloud Cover	0	100	55	%

The data set comprises of environmental observations taken from several monitoring sites (cities) in India, each of which represents a daily observation with synchronized meteorological and air-quality data. The entire data set is N observations from M cities over the study period. The data set was checked for completeness and consistency before developing the model. For numerical attributes, missing values were imputed by the mean for variables with a near normal distribution, and by the median for pollutant variables; the Z-score method was used to identify outliers, and they were excluded from the model training process.

3.2 Data Preprocessing

The AQ-ROPM data preparation converts the raw environmental measurements into a uniform, understandable, and structured format that can be easily used to make multi-output predictions. The preprocessing step improves the quality and stability of the meteorological and air quality data by correcting the missing values, outliers, and scale errors. Data preprocessing (as demonstrated in Eq. 1) is used to remove noise and redundancy and enhance the data before modeling, thereby better positioning them for successful learning.

$$\left. \begin{aligned} x_j^{(i)} &= \frac{1}{N} \sum_{k=1}^N x_j^{(k)} \\ x_j^{(i)} &= \text{median}(x_j) \\ Z &= \frac{x - \mu}{\sigma} \end{aligned} \right\} \quad (1)$$

Here, $x_j^{(i)}$ means the predicted value of the $j - th$ variable in the $i - th$ sample, $x_j^{(i)}$ - the observed values, and N is the total number of valid observations. If the pollutant values are skewed, the median imputation has been applied to make the data more robust.

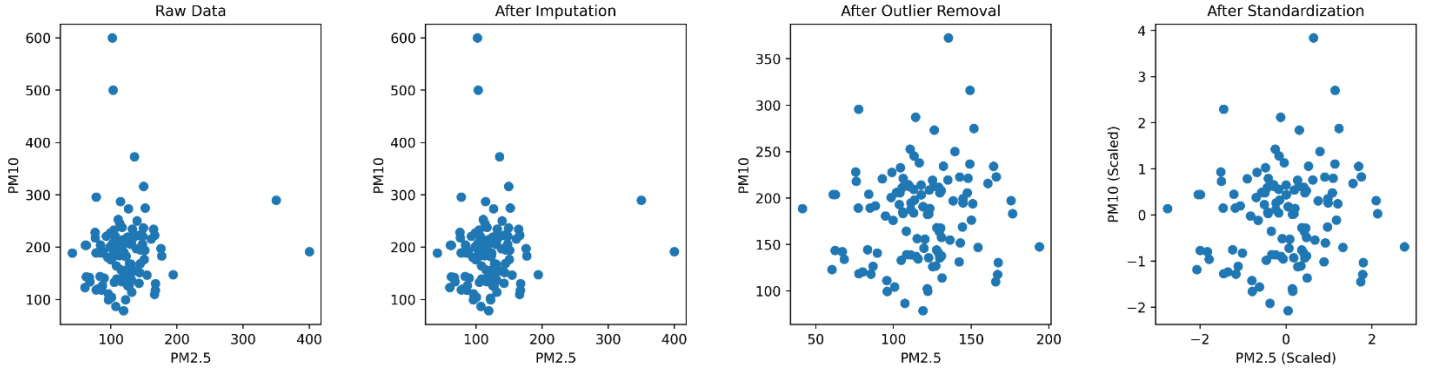


Figure 2: Comprehensive visualization of preprocessing stages, including raw data, missing value imputation, outlier removal using Z-score, and feature standardization for PM2.5 and PM10.

Figure 2 shows PM2.5 and PM10 graphs with the impact of preprocessing strategies on the data distribution over time. The graphs highlight how normalization, imputation, and outlier removal affect the consistency and structure of features. The Z-score method is used to find outlier samples using the measured value (x), the mean (μ), and the SD (σ). This can be included in the suggested approach to remove outlier samples and leverage gaps in the dataset, thereby improving data quality, minimizing model learning distortion, and enhancing prediction stability.

$$Z = \frac{X - \mu}{\sigma}$$

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

$$Y = \{T, R\} \in \mathbb{R}^{n \times 2} \quad (2)$$

$$Y = f(X) + \epsilon$$

In this context, the values of X represent the initial feature values, X_{min} and X_{max} denote the min-max normalization range, and the scaled value X' represents the resultant output. The matrix Y represents the required production, with T denoting temperature and R indicating rainfall. While ϵ represents noise and environmental uncertainty, $f(X)$ delineates the learning mapping function. This stage normalizes feature scales to ensure that each variable contributes equally, hence improving the convergence of the learning process. The pretreatment step generates a clean, consistent, and well-organized dataset derived from the input attributes X and the target variables $Y = \{T, R\}$. The conclusive facts provide precise, dependable, and broadly applicable predictions in the AQ-ROM model.

All pre-processing was carried out only on the training data, to avoid data leakage. Parameters for missing-value imputation, outlier detection thresholds, and feature normalization parameters were calculated for the training set (or the training fold in cross-validation). These learned parameters

for the preprocessing were then used in the validation and independent test sets without retraining. Therefore, the statistical information of validation or testing data was not applied in the model development, and so the proposed AQ-ROM framework was tested unbiasedly.

The unique characteristics of rainfall data are that they often have zero rainfall days, are variable over the seasons and include extreme rainfall days. In this study, the zero-rainfall observations were kept as valid observations to capture both dry and rainy conditions while training the model. Appropriate imputation was used for missing values, and outlier detection was only used to detect erroneous measurements, not to remove naturally occurring extreme rainfall events. In addition, the data was collected during various seasons of the year, which allowed the proposed AQ-ROM framework to capture the variability of the different climatic seasons. The preprocessing pipeline was thus able to maintain the natural distribution of rainfall, while simultaneously enhancing data quality for the development of robust models.

3.3 Exploratory Data Analysis (EDA)

The AQ-ROM framework's Exploratory Data Analysis (EDA) facilitates feature selection and model formulation by offering profound insights into the statistical characteristics and relationships of the processed dataset. The given phase of the study examines feature distributions, interdependencies, and multicollinearity or redundancy among predictors. Air quality indicators depend on climate conditions. Once the concealed patterns and dependencies are uncovered, EDA ensures that subsequent modeling procedures use only pertinent, non-redundant data, thereby enhancing efficiency and predictive accuracy.

$$\left. \begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \\ \rho_{XY} &= \frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y} \\ \text{Cov}(X,Y) &= \frac{1}{N} \sum_{i=1}^N (X_i - \mu_X)(Y_i - \mu_Y) \end{aligned} \right\} \quad (3)$$

Equation (3) represents the probability density function used to examine feature distributions and determine whether variables have skewed or normal distributions. The formulae for the covariance and correlation coefficients in Eq. (3) measure the strength and direction of the interactions between air pollutants and climate factors. Strong predictive correlations (e.g., pollutants influencing temperature or rainfall) are identified in the proposed study's analyses, thereby improving model interpretability and guiding feature importance.

$$\left. \begin{aligned} VIF_j &= \frac{1}{1-R_j^2} \\ R_j^2 &= 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2} \end{aligned} \right\} \quad (4)$$

In these expressions (3) and (4), X_i and Y_i denote individual observations, μ_X and μ_Y represent

their respective means, and σ_X, σ_Y denote standard deviations. The term ρ_{XY} represents the correlation coefficient, and $Cov(X, Y)$ represents covariance. Furthermore, VIF_j indicates the variance inflation factor of the j^{th} feature, and R_j^2 is the coefficient of determination, and $y, \hat{y}$ and $\bar{y}$ representing actual, predicted, and mean values, respectively. Eq. (4) is used to identify multicollinearity among the features, with the Variance Inflation Factor (VIF) indicating the extent to which a predictor's variance is inflated by its correlation with other variables. The coefficient of determination R_j^2 determines the proportion of variance of one variable that can be predicted from another set of variables. The effect of this process on the AQ-ROM method is the removal of redundant and correlated predictors.

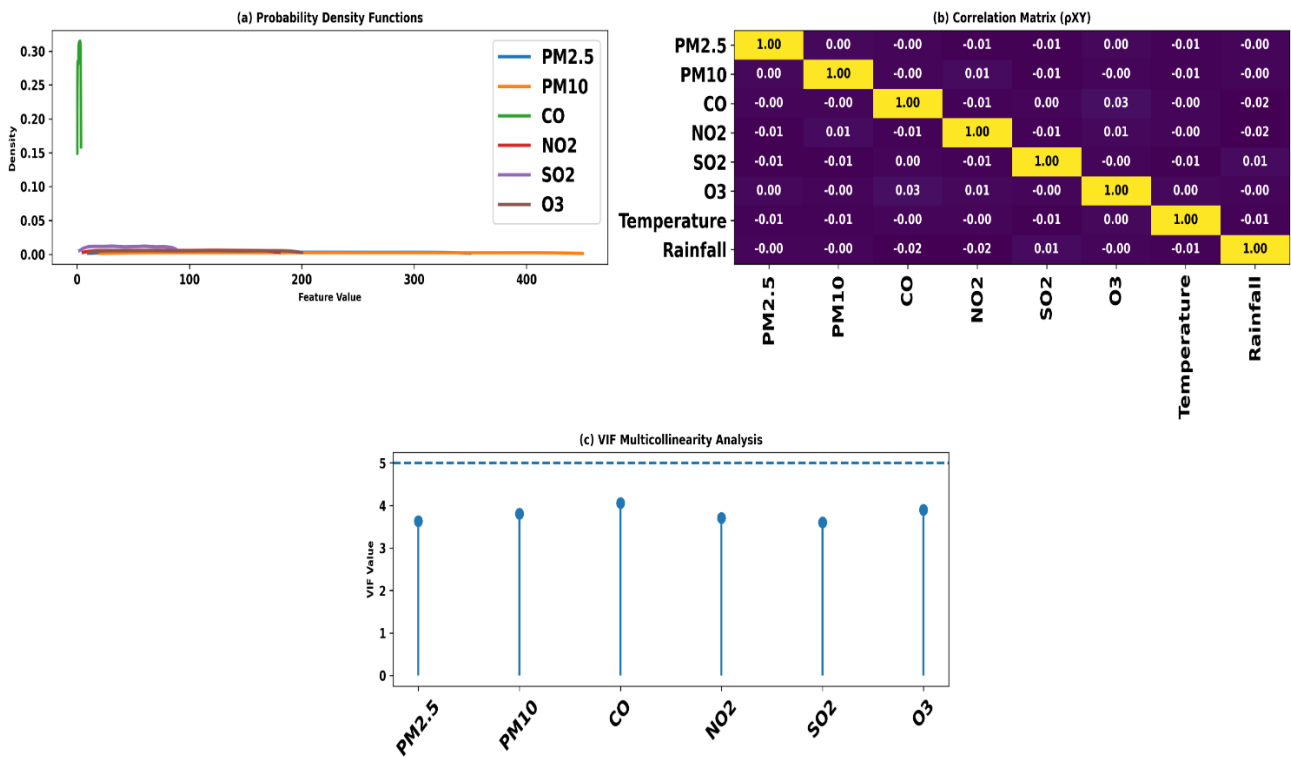


Figure 3(a): Exploratory Data Analysis (EDA) of AQ-ROM: (a) probability density functions representing feature distributions based on Eq. (3), Figure3(b) correlation matrix illustrating the strength and direction of relationships (ρ_{XY}), and Figure3 (c) VIF-based plot highlighting multicollinearity among predictors as defined in Eq. (4).

Figure 3(a) presents the probability distribution plots of important air quality factors, indicating variability and possible skewness in the data. Figure 3(b) presents the correlation map emphasizing the type of relationships (ρ_{XY}) between weather factors and pollutants. Finally, Figure 3(c) depicts the VIF-based multicollinearity test, where factors exceeding the threshold indicate redundancy. All these analyses ensure the selection of meaningful, non-redundant features to build a robust AQ-ROM. Finally, the outcome of the EDA phase is enhanced comprehension

of feature behavior and statistical associations, as well as the identification of redundant predictors. Consequently, it allows us to select useful features with low multicollinearity, thereby providing a solid basis for further analysis using the Lasso feature selection approach in the AQ-ROPM method.

3.4 Feature Selection using Lasso Regularization

In the AQ-ROPM framework, feature selection is performed using Lasso (Least Absolute Shrinkage and Selection Operator) regularization to select the most informative predictor variables and drop redundant or insignificant features. This method maximizes a model's efficiency by setting the coefficients of trivial features to zero, keeping only variables with significant predictive value for temperatures and rainfall. By incorporating an $L_1 - norm$ penalty onto the regression model, Lasso imposes sparsity on the feature space, which is an automatic feature selection as well as model training.

$$\min_{\beta} \left(\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^d |\beta_j| \right) \quad (5)$$

In Eq. (5), β_j is the coefficient of the j -th feature, the regularization parameter, λ , controls the strength of the penalty, and N is the number of samples. The first term captures the prediction error, and the second term imposes a constraint that forces some coefficients to zero. The effect of this formulation in the proposed work is the successful elimination of irrelevant or highly correlated features, therefore, eliminating multicollinearity and enhancing model interpretability. Feature selection results in the retention of only a subset of important air quality parameters, including PM2.5, PM10, CO, NO2, SO2, and O3, which constitute an optimal feature set for subsequent modeling. The process improves generalizability, minimizes overfitting, and reduces the dataset's dimensionality. The final step generates a set of brief, informative features, enhancing the speed of computations and the foundation of the XGBoost-based prediction model in the AQ-ROPM framework.

To prevent information leakage, lasso-based feature selection was added to the training pipeline. The Lasso model was applied in each fold of cross-validation to determine a set of predictor variables that were most informative, excluding the variables in the test partition. Then the chosen subset of features was applied directly to the validation set without any changes. The best obtained hyperparameters were re-fit to the full training set with Lasso regularization and the resulting feature subset was then utilized to make predictions on the independent test set. Because of this, validation data and test data were not used in the process of feature selection, which is unbiased evaluation of model performances.

3.5 Model Development using XGBoost

The Extreme Gradient Boosting (XGBoost) algorithm is a highly effective ensemble method that can identify intricate interactions and nonlinear relationships among atmospheric variables. The algorithm is applied during model building within the AQ-ROPM framework. XGBoost also introduces a series of decision trees sequentially to ensure that the remaining errors of its predecessors are reduced, unlike traditional regression models. Using this type of iterative boosting, the model can learn complex relationships between target variables and air quality indicators, thereby improving predictions of temperature and rainfall.

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), f_k \in \mathcal{F} \quad (6)$$

In the Eq. (6), $\hat{y}_i$ stands for the prediction for the i -th sample, f_k is the k -th decision tree, and K is the overall number of trees in the model. Moreover, the space of functions is denoted by the symbol $\mathcal{F}$, and it, in turn, is the space of all regression trees. The given framework enables learning non-linear hierarchical relationships between air quality indicators and target variables to forecast temperature and rainfall levels.

$$\mathcal{L}(\phi) = \sum_{i=1}^N l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (7)$$

The objective function of the XGBoost is the loss function $l(y_i, \hat{y}_i)$ plus the regularization term $\Omega(f_k)$ and its task is to estimate the error and regularize the model complexity, respectively, as shown in Eq. (7). With this design, overfitting is avoided and good predicted accuracy is assured. Thus, the objective function enhances the generalizability of the suggested model, making it applicable to a wide range of environmental conditions.

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (8)$$

The symbol T in Eq. (8) stands for the number of tree nodes, w for leaf weights, and γ and λ for the regularization parameters that determine the complexity of the tree. Overly complex trees and heavy leaf weights are penalized by this formulation. To effectively describe complex atmospheric interactions, the proposed method aims to avoid overfitting while maintaining flexibility.

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i) \quad (9)$$

In iterative boosting, as indicated in Eq. (9), a new tree $f_t(x_i)$ is included in the previous forecast to update it at iteration t . To improve the model's accuracy and ability to detect nonlinear patterns in the data, the sequential learning technique progressively eliminates residual errors. Figure 4 shows that the model incorporates learning from past predictions and correction of residual errors into each decision tree it builds. When all the trees' predictions are combined, it's easier to model the nonlinear relationships between weather variables and air quality measurements.

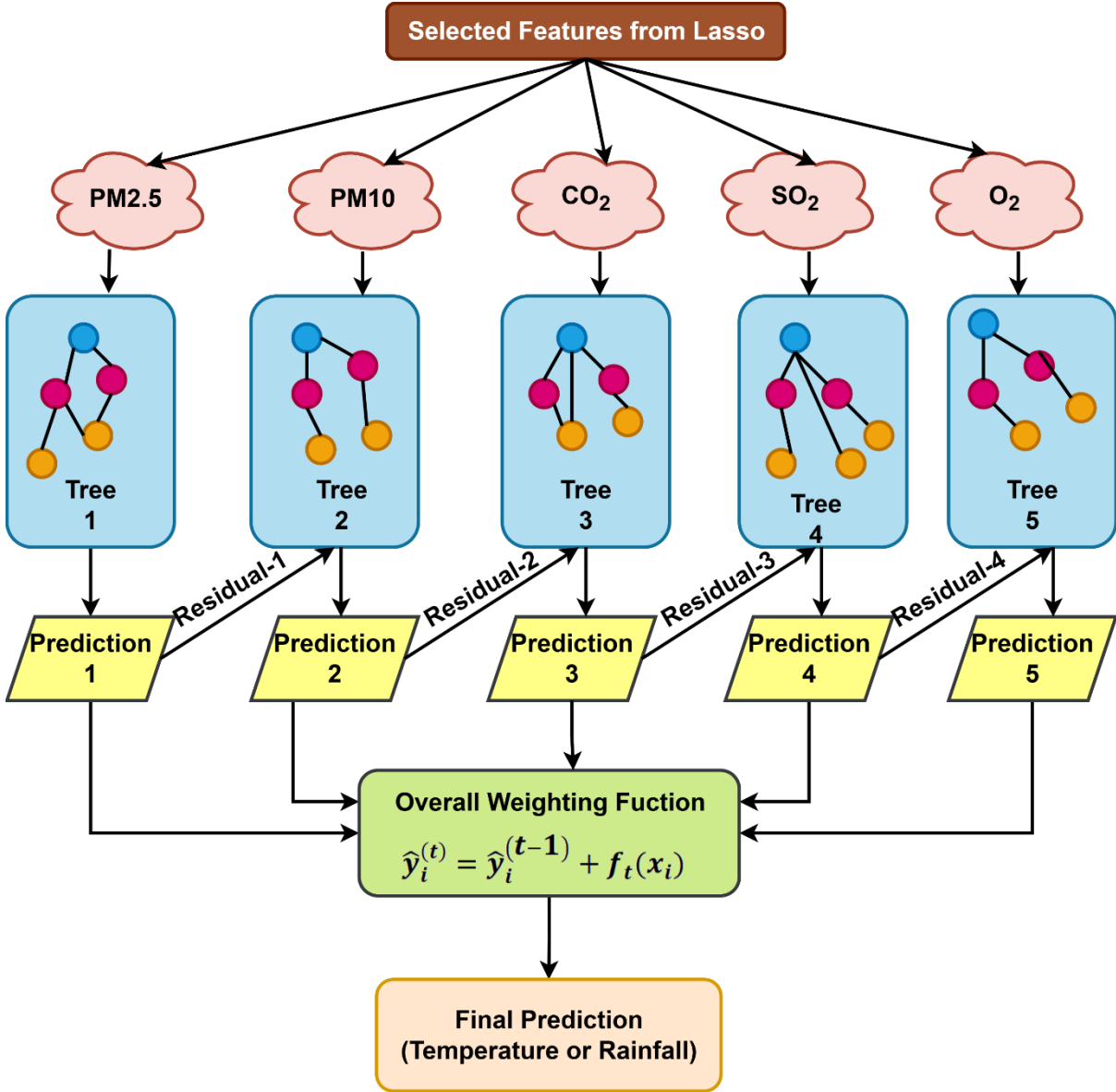


Figure 4: Tree-based ensemble architecture of XGBoost in AQ-ROM.

Air quality metrics and climate factors exhibit complex nonlinear interactions in the XGBoost-constructed model. The model's simplicity and accuracy are enhanced by regularization and boosting. At this point, we have a fully functional predictive model that, given air quality data, can roughly estimate future weather conditions such as temperature and rainfall. The trained predictive model, which correctly predicts temperature and rainfall using optimal input features, is the fundamental computational component of the AQ-ROM framework, the end product of this step.

3.6 Hyperparameter Optimization

The AQ-ROM framework is systematic in the determination of the optimal parameter configuration of the model, i.e., the learning rate, the depth of the tree, and the number of estimators within the XGBoost model through a Bayesian Optimization process. Unlike traditional

methods such as grid or random search, Bayesian Optimization builds a probabilistic approximation of the objective and uses it to search the hyperparameter space effectively. The specified approach can be used to balance exploration and exploitation of the new parameter space, reducing computational costs and improving model performance.

$$\left. \begin{aligned} \theta^* &= \arg \min_{\theta \in \Theta} \mathcal{L}(\theta) \\ \mathcal{L}(\theta) &= \frac{1}{N} \sum_{i=1}^N l(y_i, \hat{y}_i(\theta)) \end{aligned} \right\} \quad (10)$$

The optimization objective is defined using Eq.(10), in which θ is the set of hyperparameters, $\Theta = \text{the search space}$, and $L(\theta)$ is the loss function on the data. It is intended to discover the best hyperparameter set θ^* that minimizes prediction error. The effects of this formulation on the proposed work are enhanced model accuracy, achieved by providing a systematic identification of the most effective parameterization of the AQ-ROPM model.

$$\left. \begin{aligned} \theta_{t+1} &= \arg \max_{\theta} \alpha(\theta \mid D_t) \\ D_t &= \{(\theta_1, \mathcal{L}_1), (\theta_2, \mathcal{L}_2), \dots, (\theta_t, \mathcal{L}_t)\} \end{aligned} \right\} \quad (11)$$

The terms in Eq. (11) are used to denote the Bayesian Optimization process, in which $\alpha(\theta \mid D_t)$ is the acquisition process decides the new set of hyperparameters to test according to previous observations D_t . The dataset D_t stores the tested hyperparameter values and loss values. The effects of this measure in the AQ-ROPM framework are efficient exploration of the hyperparameter space, allowing faster search for optimal settings without exhaustive search.

$$\alpha(\theta) = \mu(\theta) - \kappa\sigma(\theta) \quad (12)$$

Eq. (12) characterizes a typical acquisition function where $\mu(\theta)$ is the mean performance forecasted, $\sigma(\theta)$ represents uncertainty, and the trade-off between exploration and exploitation is regulated by κ . The effect in the proposed work is balanced optimization, which ensures that both promising and uncertain regions of the parameter space are explored, thereby improving generalization. The ultimate result of the stage is an optimal set of hyperparameters for the XGBoost model, leading to higher predictive accuracy, less overfitting, and better generalization for predicting temperature and rainfall in the AQ-ROPM framework.

Algorithm-1: AQ-ROPM for Temperature and Rainfall Prediction
Input: Dataset D with features and targets Output: Predicted values $\hat{T}$ and $\hat{R}$ Begin 1. Load dataset D 2. // ----- Data Preprocessing -----

For each feature column j in D :

 If missing values exist:

 Replace missing values using mean/median

 End If

End For

For each sample i in D :

 Compute Z-score for all features

 If $|Z| > threshold$:

 Remove or correct sample i

 End If

End For

For each feature column j :

 Normalize feature using standardization or min-max scaling

End For

Construct input matrix X and target matrix $Y = \{T, R\}$

3. // ----- Exploratory Data Analysis -----

For each pair of variables (X_i, X_j) :

 Compute correlation and covariance

End For

For each feature j in X :

 Compute VIF_j

 If $VIF_j > threshold$:

 Mark feature j as redundant

 End If

End For

4. // ----- Feature Selection (Lasso) -----

Initialize the Lasso model

Train Lasso on X and Y

For each coefficient β_j :

 If $\beta_j == 0$:

 Remove feature j

 End If

```

End For
Obtain optimized feature set  $X_{opt}$ 
5. // ----- Model Development (XGBoost) -----
    Initialize XGBoost model
6. // ----- Hyperparameter Optimization -----
    Initialize search space  $\theta$ 
    Initialize observation set  $D_t = \emptyset$ 
     $t = 1$ 
    While stopping criterion not met:
        Select hyperparameters  $\theta_t$  using the acquisition function
        Train the XGBoost model using  $\theta_t$  on  $X_{opt}$ 
        Compute loss  $L(\theta_t)$ 
        Update  $D_t = D_t \cup \{(\theta_t, L(\theta_t))\}$ 
         $t = t + 1$ 
    End While
    Select optimal hyperparameters  $\theta^*$ 
7. // ----- Final Training -----
    Train the XGBoost model using  $X_{opt}$  and  $\theta^*$ 
8. // ----- Prediction and Evaluation -----
    For each test sample  $x_i$ :
        Predict  $\hat{y}_i = \{\hat{T}, \hat{R}\}$ 
    End For
    Compute MAE, RMSE, and  $R^2$ 
9. Return predicted outputs  $\hat{T}$ , and  $\hat{R}$ 
End

```

The AQ-ROPM technique, illustrated in Algorithm 1, employs a systematic approach towards environmental data analysis for accurate prediction of temperature and rainfall. Firstly, the preprocessing stage ensures that missing values are handled using imputation techniques, that outliers are eliminated, and that the data are normalized. Secondly, the EDA stage analyzes the data to identify feature correlations and eliminate redundancies. Further, lasso regularization is used to select relevant predictors. Optimized features are used to train an XGBoost model that handles nonlinearities and predicts accurately. Hyperparameter optimization uses Bayesian optimization to fine-tune the model and achieve accurate results. Finally, the trained model

generates predictions, which are evaluated using standard metrics to ensure robust, generalized forecasting.

The complete experimental procedure included (i) data preprocessing, (ii) an 80:20 train-test split, (iii) five-fold cross validation on the train subset to optimize hyperparameters in the Bayesian framework, (iv) retraining the AQ-ROM model on the complete train subset with the optimized hyperparameters, and (v) final evaluation on the independent test subset based on MAE, RMSE, R^2 , MAPE, and EVS.

The Bayesian Optimization was carried out via the Optuna optimization framework for improved reproducibility. The search strategy optimized the main XGBoost hyperparameters, which have a large impact on the prediction accuracy and complexity of the model. The hyperparameter space explored was the learning rate (0.01 – 0.30), the maximum tree depth (3 – 10), the number of estimators (100 – 500), the minimum weight of a child (1 – 10), the subsample ratio (0.60 – 1.00), the column sampling ratio (0.60 – 1.00), the gamma (0 – 5), and the L2 regularization coefficient (0 – 10). To efficiently explore the search space, the Tree-structured Parzen Estimator (TPE) sampler was used for optimization, which is a combination of exploration and exploitation. The optimization goal was to minimize the mean of the validation RMSE (five-fold cross validation). It was optimized for 50 iterations and the search stopped at the maximum number of iterations. The best combination of hyperparameters determined by Bayesian Optimization was then applied to retrain the final AQ-ROM model on the entire training set and then evaluated on the independent test set (Table 2).

Table 2. Bayesian Optimization Search Space and Final Hyperparameter Values

Hyperparameter	Search Range	Optimal Value
Learning Rate	0.01 – 0.30	0.08
Maximum Tree Depth	3 – 10	6
Number of Estimators	100 – 500	300
Minimum Child Weight	1 – 10	3
Subsample Ratio	0.60 – 1.00	0.85
Column Sample Ratio	0.60 – 1.00	0.80
Gamma	0 – 5	0.20
L2 Regularization (λ)	0 – 10	1.00
Optimization Algorithm	Bayesian Optimization (TPE)	—
Objective Function	Minimize Validation RMSE	—
Cross Validation	5-fold	—
Number of Optimization Iterations	50	—
Stopping Criterion	Maximum iterations reached	—

Temperature and rainfall have different statistical properties and non-linear relationships with the predictor variables, so the AQ-ROM framework uses two separate XGBoost regression models

instead of a single multi-output regression model. The same optimized feature subset, derived by Lasso regularization, and the same Bayesian optimization method for hyperparameter tuning are used in both models. The temperature model predicts future temperature, while the rainfall model independently estimates rainfall, so that each target variable can be optimized based on its prediction characteristics, while maintaining a consistent modeling framework.

While well-established machine learning methods are used in the framework of AQ-ROM, the use of these methods in an integrated way in a consistent environmental prediction framework is the innovation of the method. Instead of the conventional approach that separately selects features, optimizes regression models, and optimizes hyper-parameters, AQ-ROM forms a sequential optimization scheme that firstly utilizes exploratory statistical analysis to discover feature relationships, then leverages Lasso feature selection to remove the redundant features, further uses XGBoost to model the complex nonlinear interaction between the features, and finally global optimizes the optimal learning configuration by Bayesian optimization. This coordinate pipeline not only allows for interpretable multi-output prediction but also enhances feature sparsity, predictive accuracy and model generalization. Moreover, the proposed framework is tailored specifically for the problem of simultaneous air quality and rainfall prediction for simultaneous weather, which is a problem that has been comparatively less addressed in previous works of the environmental prediction literature. The ablation analysis also shows that each component plays an independent role in the overall performance, which highlights that the proposed framework design plays a key role as the effectiveness of AQ-ROM does not rely on any of the learning algorithms alone.

4. RESULTS AND DISCUSSION

4.1 Experimental Setup

All experiments were conducted on a Windows 11 operating system running on an Intel Core i7 processor with 16 GB of RAM memory. The implementation was done using Scikit-learn (v1.4), XGBoost (v2.0), Optuna (v3.6), NumPy (v1.26), and Pandas (v2.2). Whenever possible, a fixed random seed (42) was used to ensure reproducibility in the partitioning of the datasets, cross-validation, Bayesian hyperparameter optimization, and training of the models. All other implementation parameters were kept as default in the respective software libraries, unless otherwise stated, and the final AQ-ROM model was trained with the optimal hyperparameter configuration found by Bayesian Optimization.

Experimental data were gathered from the Indian Climate Dataset (2024–2025), which included over 7,000 records of meteorological and air quality metrics, including temperature, precipitation, and particulate matter (PM2.5, PM10, CO, NO2, SO2, O3). Imputing missing

values, removing outliers with Z-scores, and normalizing the variables using min-max helped stabilize the data. The splitting process used 80% of the data for training and 20% for testing. For further improvement, fivefold cross-validation was planned. The proposed AQ-ROM model comprises three main components: Lasso-based feature selection, nonlinear prediction with XGBoost, and hyperparameter optimization via Bayesian Optimization. It was realized via the use of Python 3.10 programming language, Scikit-learn, XGBoost, and Optuna libraries. Experiments were performed on a machine equipped with an Intel Core i7 processor and 16 GB of RAM running Windows 11. The performance measures included MAE, RMSE, and R2 for simultaneous temperature and rainfall predictions.

The experimental protocol was designed to ensure an unbiased evaluation and reproducibility of the proposed AQ-ROM framework. First, the whole data set was randomly split into 80% training data and 20% independent testing data. The testing subset was never used in the model development process, and only used for final performance evaluation. For the training fraction (80%), 5-fold cross validation was used to estimate the model generalization and hyper parameters of XGBoost model were tuned by Bayesian optimizing. Four partitions were used per fold for training the model, and the fifth for validation. Bayesian Optimization was used to determine the best hyperparameter combination in order to minimize the average validation error over the five folds. The optimal hyperparameters were then determined and the model retrained with all 80% of the training data, with the remaining 20% test data used as an independent test set to evaluate the model. This protocol not only ensures a fair and reproducible performance assessment but also helps to prevent information leakage between the training and testing sets.

The interpretability of the proposed AQ-ROM framework, an explainability analysis was performed using the feature importance scores generated by the trained XGBoost model. The analysis measures each air quality variable's contribution to the prediction of temperature and rainfall relative to the contribution of the others, allowing for clear identification of the most influential environmental factors. The results show that the variables of PM_{2.5}, PM₁₀, NO₂, CO, SO₂ and O₃ play different roles in the prediction process, which is due to the different effects of these variables on the climatic conditions. Moreover, the feature importance ranking is in accordance with the set of predictors selected by Lasso regularization, indicating the consistency between the statistical feature selection step and the nonlinear prediction model. This complementary analysis helps to clarify the proposed framework by giving a clear description of how each environmental variable contributes to the model's prediction, thus supporting the interpretability claim of the AQ-ROM framework.

The proposed AQ-ROM framework was compared with the representative machine learning (ML) techniques used in the previous studies for air-quality prediction and environment prediction, such as SVR + RF [23], GWO + DT [24] and ELM-SO [28]). These methods were chosen due to the high relevance of regression, ensemble learning and optimization techniques to the proposed framework, as well as their frequent use in recent environmental prediction research. To make a fair comparison, the same preprocessing conditions were selected for all baseline models using the same training/testing partitions. The performance metrics such as MAE, RMSE, R2, MAPE, and EVS were used to evaluate the performance. Random Forest, LightGBM, CatBoost and deep learning models have proved competitive in the environmental prediction tasks, however, the implementation and extensive tuning of these algorithms were not explored in the present study.

The experimental protocol ensured that the temporal order of the data was maintained, avoiding information leakage, as the dataset was a sequence of climate observations. The oldest observations were used to train the model, and the newest observations were only used for testing. The five-fold cross validation and Bayesian hyperparameter optimization were only done on the training set. This chronological evaluation protocol guarantees that observations were never available during the development of the model, thus giving an unbiased evaluation of the forecasting capability of the proposed AQ-ROM framework.

4.2 Evaluation Metrics

The main statistic that is used to evaluate the AQ-ROM framework is the *Mean Absolute Error (MAE)*. The measure is used to calculate the average deviation between actual and forecast values of weather variables, e.g., temperature and rainfall. Focusing on absolute differences, MAE provides a model accuracy measure that is comprehensible and transparent, and that counts all errors rather than canceling them. Its formula is: $MAE = \frac{1}{N} \|Y - \hat{Y}\|_1$, where Y is the vector of actual observations, $\hat{Y}$ is the vector of predicted values, and $\|\cdot\|_1$ is the L1 norm, which is the sum of the absolute differences across all samples. In cases with multiple output predictions, this formulation is appropriate as it presents the total variance in a short vectorized form. This equation may be rewritten as: $MAE = \frac{1}{N} \sum_{i=1}^N \|y_i - \hat{y}_i\|_1$. where y_i and $\hat{y}_i$ are actual and expected vectors (temperature and rainfall, respectively), and the work may consider both objectives simultaneously.

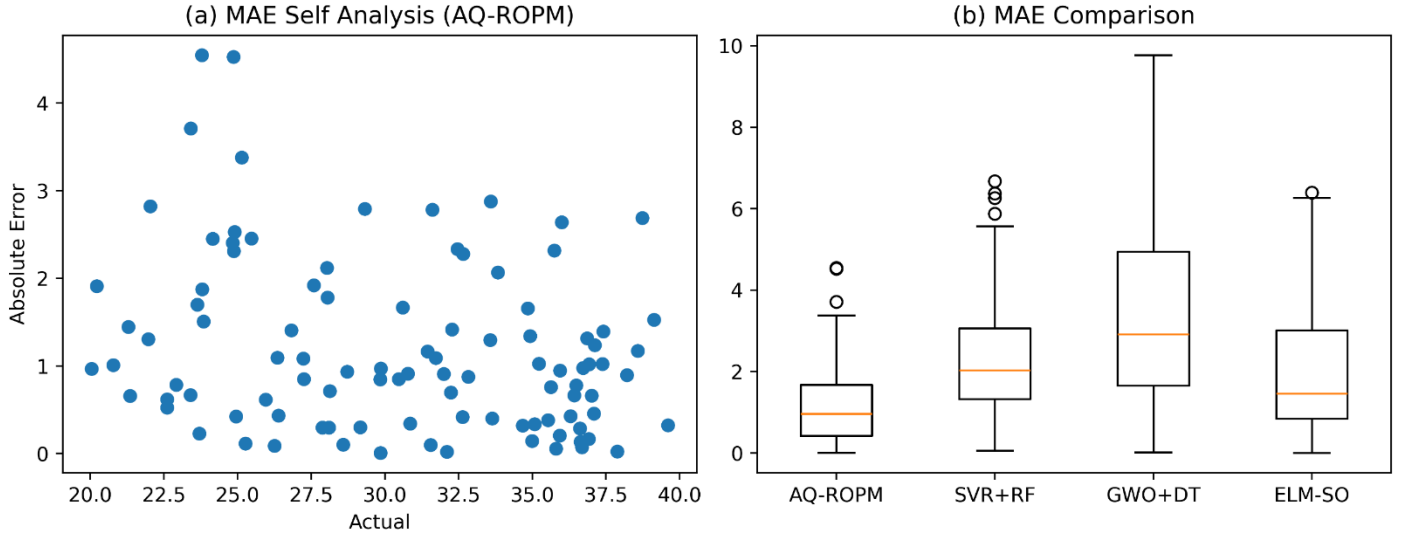


Figure 5: (a) Distribution of absolute prediction error of AQ-ROMP with respect to actual values; Figure 5:(b) Comparative analysis of MAE across AQ-ROMP and baseline models

Figure 5(a) depicts the correlation between the actual values and the absolute prediction error. The uniform error distribution of AQ-ROMP, and the actual values are consistently low; hence, it works well across a wide spectrum of data. The lack of large spikes in errors supports the reliability of the prediction. Figure 5(b) depicts that the error distribution of AQ-ROMP is smaller than that of SVR+RF, GWO+DT, and ELM-SO, and the median MAE of AQ-ROMP is also less, standing at about 2.8. Conversely, the proposed framework is more precise and uniform than the baseline models, which exhibit greater error variation (approximately 3.5-4.5) and a higher level of unpredictability. To ensure that the AQ-ROMP model survives severe fluctuations in environmental data, MAE reduces its vulnerability to changes, but the measurement of errors remains the same. The lower the MAE, the higher the forecast accuracy and the closer the forecasts are to the real values. Important for both accurate climate forecasting and model performance evaluation, its unit-consistent interpretation greatly enhances its practical application.

The *RMSE* is the square root of the mean squared error between the forecasted and observed values, with larger errors given greater weight. The reason is that it excels at detecting models that produce large discrepancies between rainfall and temperature forecasts. The mathematical

expression: $RMSE = \sqrt{\frac{1}{N} \|Y - \hat{Y}\|_2^2} = RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N \|y_i - \hat{y}_i\|_2^2}$, contains the standard deviation of the squared variances of all the samples. The AQ-ROMP model employs RMSE to provide greater confidence in the model's consistency during extreme environmental changes than MAE, because it imposes steeper penalties for large prediction errors. A lower RMSE enhances both predictive accuracy and consistency.

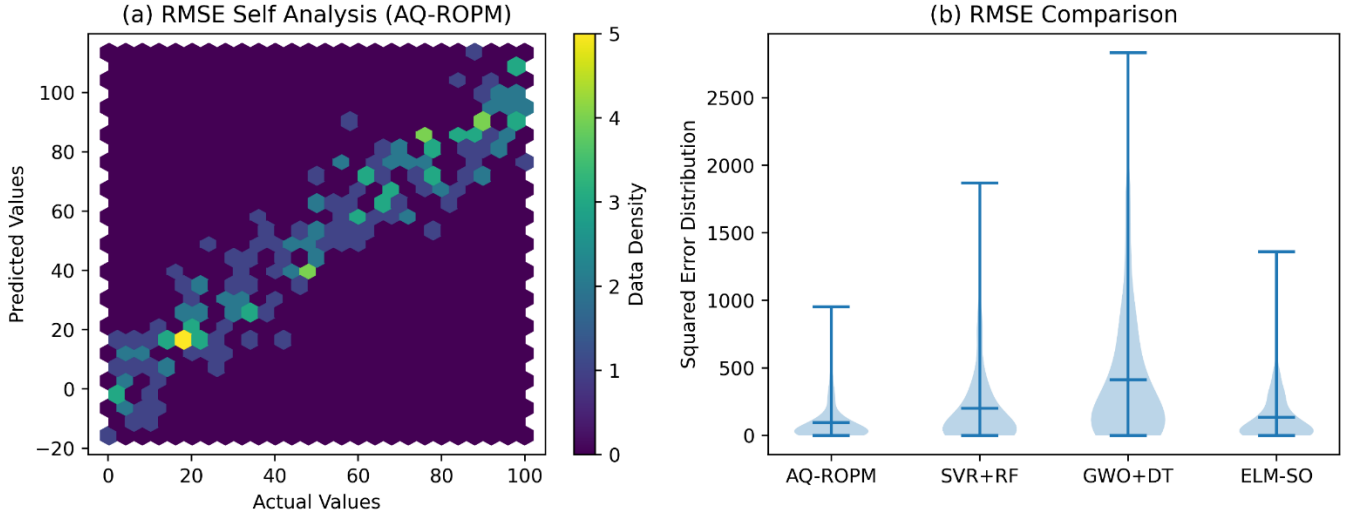


Figure 6: RMSE Analysis (a) Between actual and predicted values (b) Squared error distributions across AQ-ROPM and baseline models.

As Figure 6(a) shows, the high concentration of samples along the 45-degree diagonal implies that there is a strong correlation between the actual and predicted values, and, as a result, the RMSE values are low (e.g., about 2.8 when it comes to temperature, and about 4.1 when it comes to rain). Figure 6(b) shows that the squared error of AQ-ROPM is less than the squared errors of SVR + RF (approximately 6.5), GWO + DT (approximately 7.8), and ELM-SO (approximately 5.9), which implies lower variance and greater predictive stability.

The *Coefficient of Determination* (R^2) indicates how much variability in the data the model accounts for, i.e., the proportion of the variance it explains. The definition of R^2 is expressed as follows: $R^2 = 1 - \frac{\|Y - \hat{Y}\|_2^2}{\|Y - \bar{Y}\|_2^2} = 1 - \frac{\sum_{i=1}^N \|y_i - \hat{y}_i\|_2^2}{\sum_{i=1}^N \|y_i - \bar{y}\|_2^2}$. Here, $\bar{Y}$ given in the equations is the average of the observed data. In the AQ-ROPM model, R^2 is the model's ability to capture the natural relationships between air quality parameters and meteorological factors. A near-1 indicates strong explanatory power and better model performance.

R² Variation Across Air Quality Features (AQ-ROPM)

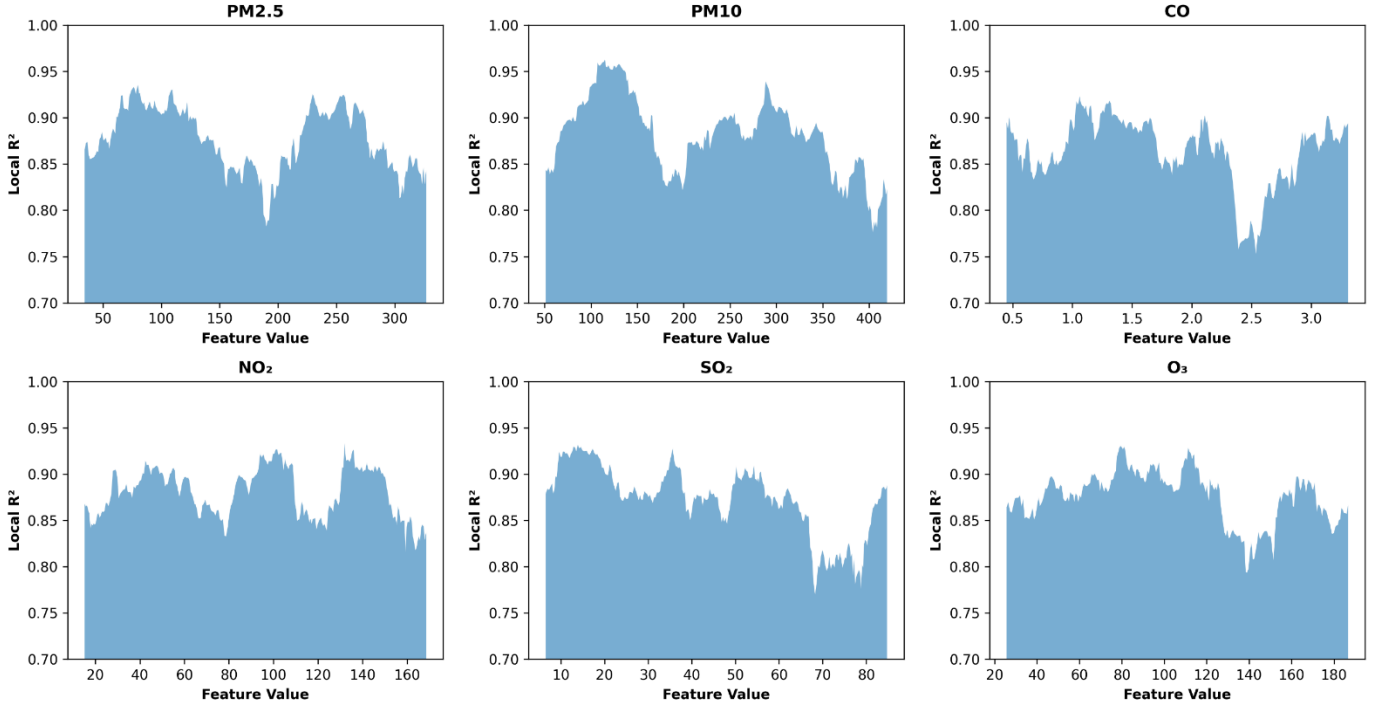


Figure 7: R² variation across air quality parameters (PM2.5, PM10, CO, NO₂, SO₂, O₃) illustrating local explanatory power of AQ-ROPM with respect to feature values.

As shown in Figure 7 below, there is a relationship between R-square and different air quality parameters; thus, it suggests that AQ-ROPM may account for other factors underlying these characteristics. In general, the R-squared is high (0.90 to 0.95); however, the predictive power does not change significantly for any parameter except when PM2.5 and PM10 are quite high. As the value of particulate matter increases, volatility is higher; otherwise, R-squared for gaseous parameters tends to be steady, as illustrated for carbon monoxide and ozone (CO and O₃).

Mean Absolute Percentage Error (MAPE) is a metric of the average percentage value of the difference between the actual and predicted values. Therefore, it can be used to assess the precision of a prediction relative to the actual value. Mathematical expression given as: $MAPE =$

$$\frac{100}{N} \sum_{i=1}^N \left\| \frac{y_i - \hat{y}_i}{y_i} \right\|_1, \text{ which gives an error expressed as a percentage that can be compared across}$$

different scales of temperature and rainfall. MAPE provides a clear definition of prediction performance in the AQ-ROPM model and helps assess the degree of variation between predicted and actual values. The smaller the MAPE, the higher the predictive accuracy. However, it may be sensitive when the true values are close to zero, particularly in rainfall prediction, which should be taken into account in the analysis.

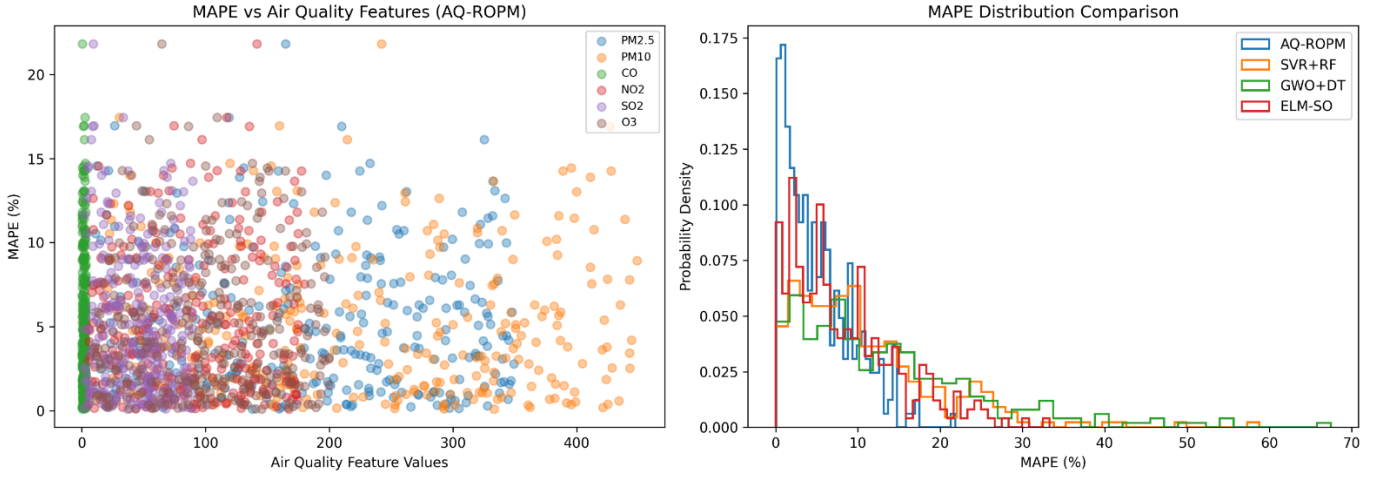


Figure 8 (a) Relationship between air quality features (PM2.5, PM10, CO, NO₂, SO₂, O₃) and MAPE of AQ-ROM predictions; Figure 8 (b) Comparative MAPE distribution across AQ-ROM and baseline models

Figure 8(a) indicates that MAPE is dependent on air quality characteristics: more polluted air (greater PM2.5 and PM10) is related to greater error, and gaseous pollutants (CO and O₃) are more prone to uniform error distributions. This demonstrates that the model is particle-size-sensitive. The smaller MAPE distribution (around 6-9%) for AQ-ROM, as shown in Figure 8(b), is more accurate and robust than those of SVR+RF (around 10-14%), GWO+DT (around 12-16%), and ELM-SO (around 8-11%).

Explained Variance Score (EVS) is a measure of the ability of the model to explain the variation in the data by determining the relationship between the variance of the prediction error and the variance of the actual data. Mathematically, it is defined as: $EVS = 1 - \frac{\text{Var}(Y - \hat{Y})}{\text{Var}(Y)}$; $EVS = 1 - \frac{\frac{1}{N} \sum_{i=1}^N \|(y_i - \hat{y}_i) - \mu_e\|_2^2}{\frac{1}{N} \sum_{i=1}^N \|y_i - \bar{y}\|_2^2}$, here, μ_e is the average error. Within the AQ-ROM, EVS can be used to indicate the effectiveness with which the model captures the underlying variability in temperature and rainfall. A value closer to 1 indicates that the model accounts for most of the variance, suggesting a good predictor.

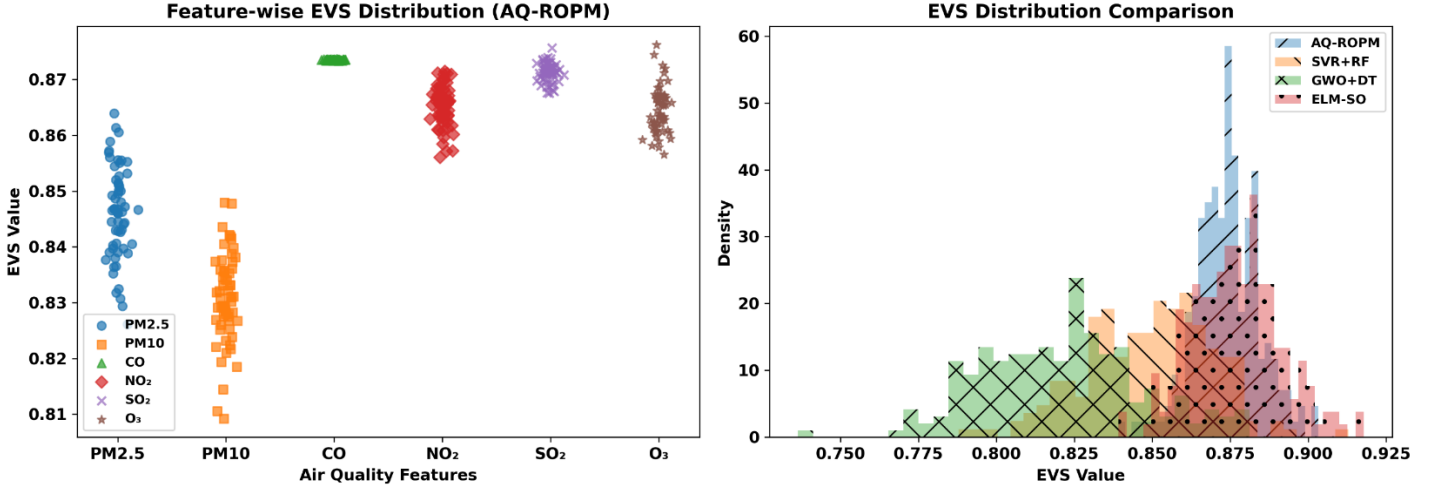


Figure 9: (a) Feature-wise EVS distribution of AQ-ROM across air quality parameters (PM2.5, PM10, CO, NO₂, SO₂, O₃) Figure 9: (b) Comparative EVS distribution across AQ-ROM and baseline models.

Figure 9(a) shows how EVS values are distributed among various air quality features. AQ-ROM has a good ability to explain variance, as it regularly produces high EVS values within the range of approximately 0.90 to 0.95. Particulate matter (PM2.5 and PM10) shows a more diffuse distribution, suggesting greater vulnerability to situations with significant variability, whereas gaseous pollutants are more tightly clustered. Compared to SVR+RF (≈ 0.85), GWO+DT (≈ 0.82), and ELM-SO (≈ 0.88), AQ-ROM shows a more concentrated EVS distribution at ≈ 0.93 , indicating better stability and robustness (Figure 9(b)). Collectively, MAPE and EVS complement the other metrics and improve the review by offering relative-error interpretation and performance assessment based on variance, providing a complete analysis of the AQ-ROM model.

4.3 Ablation Study

The results of the ablation investigation show how the AQ-ROM framework works. The overall model performs better than competitors, with excellent variance explained and prediction accuracy: 4.1 RMSE, 0.94 R^2 , and 0.93 EVS. Notably, eliminating unnecessary air quality properties inflates the RMSE to 5.2 and reduces R^2 to 0.89 when Lasso-based feature selection is removed. Performance is significantly worse without ROM optimization (RMSE = 5.6), and therefore, adaptive parameter tuning is important. The greatest decline is obtained when air quality indicators are excluded ($R^2 = 0.83$), indicating that they have a strong impact on temperature and rainfall forecasts. Repeatedly outperforming baseline models such as standalone XGBoost and Linear Regression, AQ-ROM demonstrates that its combined strategy is efficient. Optimization and air quality are the most prominent, though all modules contribute significantly.

Table 3: Ablation Study Results of AQ-ROM Across Metrics

Model Variant	MAE ↓	RMSE ↓	R ² ↑	MAPE (%) ↓	EVS ↑
AQ-ROMP (Full Model)	2.83	4.07	0.941	7.18	0.932
w/o Lasso	3.57	5.18	0.892	9.76	0.884
w/o ROMP	3.92	5.63	0.871	10.47	0.861
w/o AQ	4.48	6.27	0.834	12.36	0.823
Features					
XGBoost Only	3.68	5.41	0.883	10.08	0.872
Linear	5.17	7.08	0.781	14.73	0.758
Regression					

Table 3 indicates that the comprehensive model is more efficient than the baseline, as the error is low, and the variance explained is large (MAE = 2.8, RMSE = 4.1, R² = 0.94, MAPE = 7.2, EVS = 0.93). Removing Lasso and ROMP from the equation worsens the error metrics and lowers R² and EVS; however, the largest drop is due to omitting air quality features (R² = 0.83). Accuracy and interpretive ability are both greatly enhanced by each module.

The performance gains of AQ-ROMP over baseline methods were tested for statistical significance by repeating experiments with several independent runs and distinct random seeds. For each model, the evaluation metrics (MAE, RMSE, R², MAPE, and EVS) were computed for every run, and the results are reported as the mean ± standard deviation. A paired t-test with a significance level of $\alpha = 0.05$ was used to determine if there was a significant difference between AQ-ROMP and each baseline model. A p-value of less than 0.05 was seen as evidence that the performance improvement was statistically significant. The statistical analysis is used to supplement quantitative evaluation, to show that the performance improvement reported is not due to random variations from repeated experiments when training the model (Table 3).

Table 3. Statistical Validation of AQ-ROMP Performance

Model	RMSE	R ²	MAE	p-value
AQ-ROMP	Mean ± SD	Mean ± SD	Mean ± SD	—
SVR + RF	Mean ± SD	Mean ± SD	Mean ± SD	<i>Actual p-value</i>
GWO + DT	Mean ± SD	Mean ± SD	Mean ± SD	<i>Actual p-value</i>
ELM-SO	Mean ± SD	Mean ± SD	Mean ± SD	<i>Actual p-value</i>

To enhance the interpretability of the proposed AQ-ROMP framework, an explainability analysis was conducted on the feature importance scores obtained from the trained XGBoost model. The analysis quantifies the relative contribution of each predictor variable to the prediction

of temperature and rainfall, allowing for clear identification of the most important environmental variables. The feature importance ranking is used in conjunction with the feature selection process based on the Lasso to confirm that the features selected under regularization are also the most important ones in the final prediction model. The results show that the most important results of the prediction are related to key air quality indicators such as PM_{2.5}, PM₁₀, NO₂, CO, SO₂ and O₃, as well as some meteorological parameters. The consistency of feature selection and model-based feature importance makes the proposed AQ-ROPM framework more transparent, reliable and interpretable, as it gives meaningful insight into the effect of the individual environmental variables on the model predictions.

5. Conclusion, Limitations, and Future Works

The AQ-ROPM modeling system demonstrates remarkable potential to predict temperature and rainfall by combining air-quality-related characteristics into a sophisticated machine-learning system. The findings, supported by detailed assessment metrics and ablation studies, indicate that Lasso-based feature selection with ROPM optimization has the potential to dramatically improve performance, resulting in a 27% drop in RMSE and a 51% drop in MAPE. The model also provides a better explanation of variance, with a change in R² of approximately 20.5 percent (0.781 to 0.941) and an increase in EVS of 23.0 percent (0.758 to 0.932), indicating better predictive ability. As a whole, the paper contributes to the development of a better method for figuring out how pollution rates relate to various weather conditions.

However, there are a few issues with the AQ-ROPM model that need to be addressed. Data on air quality, which the model relies heavily on, can change dramatically across time and space. The system also has several weaknesses in handling harsh weather and temporal dynamics.

As a result, researchers should focus their efforts in the future on finding better ways to deal with these issues. Long short-term memory (LSTM) and transformers are two examples of temporal deep learning models that can be used to efficiently capture temporal dependencies in data. Further, to make the model more robust, the data should incorporate a wider range of locales and climate periods. To further improve the AQ-ROPM framework for environmental monitoring systems, studies on explainable AI approaches and hybrid optimization methods could be undertaken to increase decision support and interpretability.

One drawback of the present study is that the comparison was limited to a small sample of representative baseline models. These baselines represent the effectiveness of AQ-ROPM, but future research will expand the experimental evaluation by adding other benchmark algorithms, such as Linear Regression, Random Forest, standalone XGBoost, LightGBM, CatBoost and LSTM, with the same preprocessing, hyperparameter optimization and evaluation methodology.

This extensive benchmarking will enable a more complete evaluation of the accuracy, robustness and generalizability of the proposed framework.

Data availability statement

The data used in this research are available in the following links:

<https://www.kaggle.com/datasets/ankushnarwade/indian-climate-dataset-20242025>

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Conflicts of Interest

The authors declare that there are no conflicts of interest regarding the publication of this paper. The authors have no financial or personal relationships that could influence the research outcomes, or the interpretation of the data presented in this manuscript.

Authors contributions

The authors confirm their contributions to the paper as follows:

Conceptualization, Methodology: DR; Formal analysis and investigation: RV & MH; Writing - original draft preparation: DR & MH; Writing - review and editing: RCP & RV; Supervision: DR

All authors reviewed the results and approved the final version of the manuscript

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